

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester III Examination, 2016-17

Paper No. MIB-301: Algebra (Old Course)

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. (Compulsory — 2 marks each.)

- (a) Give an example of each of the following matrices: symmetric, skew-symmetric, Hermitian, and skew-Hermitian.
- (b) If A and B are Hermitian matrices, show that $AB + BA$ is Hermitian and $AB - BA$ is skew-Hermitian.
- (c) Define the Echelon form of a matrix. What is the rank of a 4×4 matrix whose entries are all 1?
- (d) Let $*$ be the binary operation defined on the set $\mathbb{Q}_+$ of all positive rational numbers by $a * b = \frac{ab}{5}$. Find the identity element and the inverse of an element $a \in \mathbb{Q}_+$.
- (e) Show that in a group G , if $(ab)^2 = a^2b^2$ for all $a, b \in G$, then G must be abelian.
- (f) Let G be a group. If $a \in G$ is of order n , show that $o(g^{-1}ag) = n$ for any $g \in G$.
- (g) If H is a subgroup of a group G , show that $HH = H$.
- (h) Let $G = \langle a \rangle$ be a cyclic group of order 13. Find the orders of a^5 and a^9 .
- (i) Define cyclic permutation, and even and odd permutations, and give one example of each.
- (j) Show that a group of prime order is cyclic.
- (k) Let N be a normal subgroup of G and K be a subgroup of G . Show that $N \cap K$ is a normal subgroup of K .

Question 2.

- (a) Show that every square matrix can be expressed uniquely as $P + iQ$, where P and Q are Hermitian matrices. [6]
- (b) Let A be a square matrix of order n . Show that:

$$A(\text{adj } A) = (\text{adj } A)A = |A|I_n$$

where I_n is the identity matrix of order n .

[6]

Question 3.

- (a) By using elementary transformations, find the rank of the following matrix:

$$A = \begin{pmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{pmatrix}$$

[6]

- (b) Find all solutions of the following system of linear equations:

$$\begin{aligned} 3x + 4y - z - 6w &= 0 \\ 2x + 3y + 2z - 3w &= 0 \\ 2x + y - 14z - 9w &= 0 \end{aligned}$$

[6]

Question 4.

- (a) Prove that a finite semigroup G in which both the cancellation laws hold is a group. Give an example in which this result fails to hold. [6]
- (b) Prove that the union of two subgroups of a group G is a subgroup of G if and only if one is contained in the other. [6]

Question 5.

- (a) Express the permutation:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 4 & 3 & 2 & 6 & 7 & 1 & 5 \end{pmatrix}$$

as the product of disjoint cycles, determine whether it is even or odd, and find its order. [6]

- (b) Let $f : G \rightarrow G'$ be a group homomorphism. Show that f is a monomorphism (one-to-one homomorphism) if and only if $\ker f = \{e\}$, where e is the identity in G . [6]

Question 6.

- (a) Prove that the order of a cyclic group $G = \langle a \rangle$ is equal to the order of its generator a . [6]
- (b) Prove that a subgroup H of a group G is a normal subgroup of G if and only if $g^{-1}hg \in H$ for each $g \in G, h \in H$. [6]

Question 7.

- (a) Let $f : G \rightarrow G'$ be an onto group homomorphism. If H and H' are normal subgroups of G and G' respectively, show that $f(H)$ and $f^{-1}(H')$ are normal subgroups of G' and G respectively. [6]
- (b) If H and K are two subgroups of a group G such that H is normal in G , prove that:

$$(HK)/H \cong K/(H \cap K)$$

[6]